Section D — AI-Augmented Learning

Cisco Packet Tracer — Router-on-a-Stick Practical

STEP 1 · BUILD WITH AI

AI Tool Used: ChatGPT

1. Exact Prompt Given to AI

I am doing a Cisco Packet Tracer practical. Generate a complete Cisco IOS configuration for a Router-on-a-Stick topology.

Requirements:

Configure VLAN 10 and VLAN 20 on the switch.

Assign switch ports to VLAN 10 and VLAN 20.

Configure an 802.1Q trunk between the switch and router.

Configure Router-on-a-Stick sub-interfaces on the router for VLAN 10 and VLAN 20.

Configure IP addresses/gateways so that a PC in VLAN 10 can communicate with a PC in VLAN 20.

Provide complete switch and router commands.

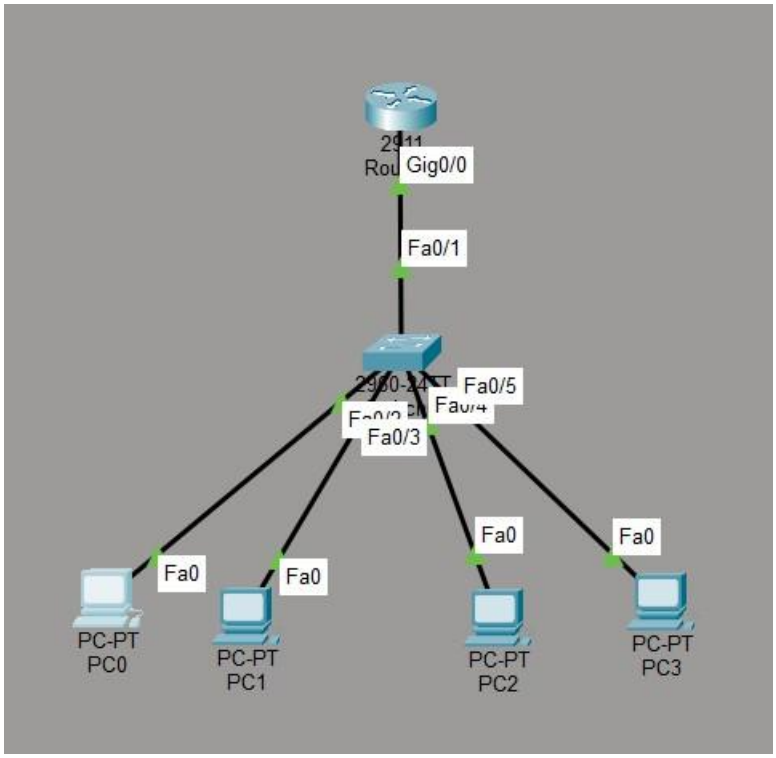
Explain how to configure the PCs.

The final configuration should allow successful ping between the VLAN 10 and VLAN 20 hosts.

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2. AI Original Configuration Script

Topology:



IP Address Plan:

Device	VLAN	IP Address	Subnet Mask	Gateway
PC0	10	192.168.10.10	255.255.255.0	192.168.10.1
PC1	20	192.168.20.10	255.255.255.0	192.168.20.1
Router Subinterface	10	192.168.10.1	255.255.255.0	—
Router Subinterface	20	192.168.20.1	255.255.255.0	—

Switch — AI Original:

```
enable configure terminal  vlan 10  name
VLAN10 exit  vlan 20  name VLAN20 exit
interface fastEthernet 0/1  switchport mode
access  switchport access vlan 10  no
shutdown exit  interface fastEthernet 0/2
switchport mode access  switchport access
vlan 20  no shutdown exit  interface
gigabitEthernet 0/1  switchport mode trunk
no shutdown exit  end write memory
```

Router — AI Original:

```
enable configure terminal  interface
gigabitEthernet 0/0  no shutdown exit
interface gigabitEthernet 0/0.10  encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0 exit
interface gigabitEthernet 0/0.20  encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0 exit  end write
memory
```

3. Testing in Packet Tracer

PC0 Configuration:

IP Address: 192.168.10.10
Subnet Mask: 255.255.255.0 Default
Gateway: 192.168.10.1

PC1 Configuration:

IP Address: 192.168.20.10
Subnet Mask: 255.255.255.0 Default
Gateway: 192.168.20.1

Test command from PC0: **ping**

192.168.20.10

STEP 2 · TEST & DEBUG (WITHOUT AI)

Example bug found during testing: the trunk must carry both VLAN 10 and VLAN 20. If the trunk is restricted to only VLAN 10, VLAN 20 traffic cannot cross the switch-router trunk.

Corrected Switch Trunk Configuration:

```
interface gigabitEthernet 0/1  switchport mode trunk
switchport trunk allowed vlan 10,20  no shutdown
exit
```

Corrected Version — Complete Switch Configuration

```
enable
configure terminal
 vlan 10  name
VLAN10 exit
vlan 20  name
VLAN20 exit
interface fastEthernet 0/1  switchport
mode access  switchport access vlan
10  no shutdown exit
interface fastEthernet 0/2  switchport
mode access  switchport access vlan
20  no shutdown exit
interface gigabitEthernet 0/1  switchport mode trunk
switchport trunk allowed vlan 10,20  no shutdown
exit  end write memory
```

Corrected Version — Router Configuration

```
enable
configure terminal

interface gigabitEthernet 0/0  no shutdown
exit
interface gigabitEthernet 0/0.10  encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0 exit
interface gigabitEthernet 0/0.20  encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0 exit  end write
memory
```

3–4 Line Debugging Note

I tested the AI-generated configuration in Cisco Packet Tracer and found that the trunk needed to carry both VLAN 10 and VLAN 20. I corrected the trunk configuration by explicitly allowing VLANs 10 and 20. After making the correction, the Router-on-a-Stick configuration worked correctly and PC0 in VLAN 10 successfully pinged PC1 in VLAN 20. This was necessary because inter-VLAN traffic must be carried through the trunk for both VLANs.